Project Title

Plagiarism Detection Engine for Academic Submissions

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

The Plagiarism Detection Engine for Academic Submissions is an intelligent system developed to detect and analyze similarities in student assignments with high accuracy and efficiency. The project aims to assist educational institutions in maintaining academic integrity by identifying copied or near-duplicate content while minimizing false detections. Initially, the system indexes a corpus of student submissions and applies Rabin–Karp fingerprinting along with the Winnowing algorithm to generate compact document fingerprints and efficiently shortlist candidate document pairs. For each shortlisted pair, a Suffix Array is constructed, and the Longest Common Prefix (LCP) Array is used to identify maximal common substrings that exceed a predefined similarity threshold. The detected matching sections are highlighted in a side-by-side comparison report, enabling users to easily examine overlapping content. The system also calculates an overall similarity score and performs false-positive analysis by comparing results against a hand-labeled dataset to evaluate detection accuracy. By combining efficient indexing techniques with advanced string-matching algorithms, the proposed solution offers a scalable, reliable, and computationally efficient plagiarism detection framework suitable for universities, colleges, and online learning platforms handling large volumes of academic submissions every semester.

Signature of the Guide